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February 4th, 2026

Dear Editor-in-Chief and Editorial Board members of *Ecology Letters*,

Please find enclosed our manuscript entitled "***Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange***", which we are pleased to submit for consideration as a letter in *Ecology Letters*.

The Great American Biotic Interchange (**GABI**) is one of the most famous biogeographic experiments that shaped modern ecosystems. It characterises the biotic admixture that occurred between the Americas, catalysed by the closure of the Isthmus of Panama in the Mio-Pliocene. A puzzling feature of the GABI is that this exchange has been **asymmetric**: a larger number of North American taxa successfully established to the South than the other way around. Despite being the scope of a wealth of study, the **underlying causes behind such an asymmetry remain unresolved**. This study envisages the asymmetry of the GABI from an innovative perspective, and proposes valuable insights into the long-debated processes that shaped this textbook example of macroecological imbalance.

Coupling high-resolution **palaeoclimatic reconstructions** covering the last 5 million years with cutting-edge spatially-explicit **eco-evolutionary simulations**, we **test** long-standing **hypotheses relating the asymmetry of the GABI to palaeoclimate dynamics**. Contrary to empirically-derived expectations, our simulations indicate that, when climate is the only constraint for species dispersal, **North America is easier to reach from South America than the other way around**. In addition, most of our simulated interchanges involved taxa from **tropical biomes** of both continents, in line with predictions of biome selectivity made from phylogenies, but not from the fossil record. This suggests that classical views of the factors shaping the asymmetry of the GABI drawing on fossils could have been affected by a taphonomic bias. Our computer experiments **challenge the paradigm** solely relating the **asymmetry of the GABI to palaeoclimate dynamics**, suggesting that other factors must have been at play to fashion the disproportionate success of North American taxa compared Southern ones during the GABI.

To our knowledge, our study is the first to provide **mechanistic evidence** to the question of the asymmetry behind the GABI. This work takes full advantage of **simulated data**, a powerful source of evidence unaffected by any upstream bias related to preservation or sampling, to provide a **new perspective to long-standing questions** in the fields of biogeography and palaeontology. It constitutes a **new step towards a better understanding of the processes underlying large-scale ecological events that shaped the biosphere**. In addition, the research approach implemented here is outstandingly new to all co-authors − myself included −, who never published on eco-evolutionary modelling. Hence, we think that our work fits the scopes of *Ecology Letters*, and hope that you will share our excitement for the study presented.

Also, for qualified reviewers, we would like to recommend:

- Dr. Alexander Skeels (alexander.skeels@anu.edu.au)
- Dr. Théo Gaboriau (Theo.Gaboriau@unil.ch)
- Dr. Roniel Freitas-Oliveira (freitasronielbio@gmail.com)
- Dr. Christine D. Bacon (christine.bacon@bioenv.gu.se)
- Dr. Juan D. Carrillo (juan.carrillo@mnhn.fr)
- Prof. Darin A. Croft (dcroft@case.edu)

Best regards,

Lucas Buffan, on behalf of my co-authors

Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange

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Statement of authorship

S.V. and L.B. designed the project. L.B. ran the simulations, performed the analyses, produced the figures and wrote the manuscript with inputs from all co-authors.

Data availability statement

R scripts for generating, analysing and visualising the data are available at https://github.com/Buffan3369/GABI-gen3sis.git. Simulation landscapes and config files are available at https://doi.org/10.6084/m9.figshare.31160704.

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